

Education

- 2025-pres. Harvard University, Cambridge, MA, PhD in Physics
- 2020-2024.5 Brown University, Providence, RI. B.Sc. in Mathematical Physics
Thesis: Reconstructing Spin Liquids
- 2014-2020 The Overlake School, Redmond, WA

Publications

- 2025 **L. Z. Brito**, J. B. Marston. *Reconstructing Spin Hamiltonians of 2D Gutzwiller-Projected Wavefunctions*, preprint. [[arXiv:2510.15034](#)].
L. Z. Brito, R. Cabeen, D. H. Laidlaw. *General Diffusion MRI Gray Matter Microstructure Factors Predict Cognitive Scores*, preprint. [forthcoming].
- 2024 **L. Z. Brito**, S. Carr, J. A. Jacoby, J. B. Marston. *Hamiltonian Reconstruction: the Correlation Matrix and Incomplete Operator Bases*, Physical Review B. [[PhysRevB.110.125118](#)].
- 2023 D. Candoli, I. K. Nikolov, **L. Z. Brito**, S. Carr, S. Sanna, V. F. Mitrović, *PULSEE: A software for the quantum simulation of an extensive set of magnetic resonance observables*, Computer Physics Communications. [[doi:j.cpc.2022.105898](#)].

Employment

- 2023-2025 *Instructional Lab Assistant, Brown Department of Physics*
Specialization in quantum optics. Designed and implemented tabletop demonstration of Bell's inequality, single photon interferometry. Wrote Python photon acquisition software for laboratory usage.
- 2022-2023 *Teaching Assistant, APMA1930W - Probabilities in Quantum Mechanics*
Wrote and lectured supplementary curriculum on quantum information theory and related topics. Substitute lectured to 30+ students.
- 2017-2018 *Assistant instructor, Play-Well Teknologies*

Awards and Recognition

- 2025-2026 Harvard Purcell Fellowship.
- 2023-2024 Goldwater Scholar in Physics and Astronomy.

Relevant Coursework

Physics	- Advanced Electromagnetic Theory - Quantum Mechanics I*, II* - Advanced Quantum Mechanics* - Advanced Statistical Mechanics* - Computational Physics*	- Quantum Theory of Fields I*, II* - Quantum Many-Body Theory* - Gauge Theory* - Standard Model* - Biophysics*
Computer Science	- Accelerated Intro to Computer Science - Data Science - Intro to Software Engineering	- IS in Biophysical Scientific Visualization - Interdisciplinary Scientific Visualization* - Deep Learning*
Mathematics	- Linear algebra - Partial Differential Equations - Abstract Algebra	- Complex Analysis - Differential Geometry* (* = graduate level)